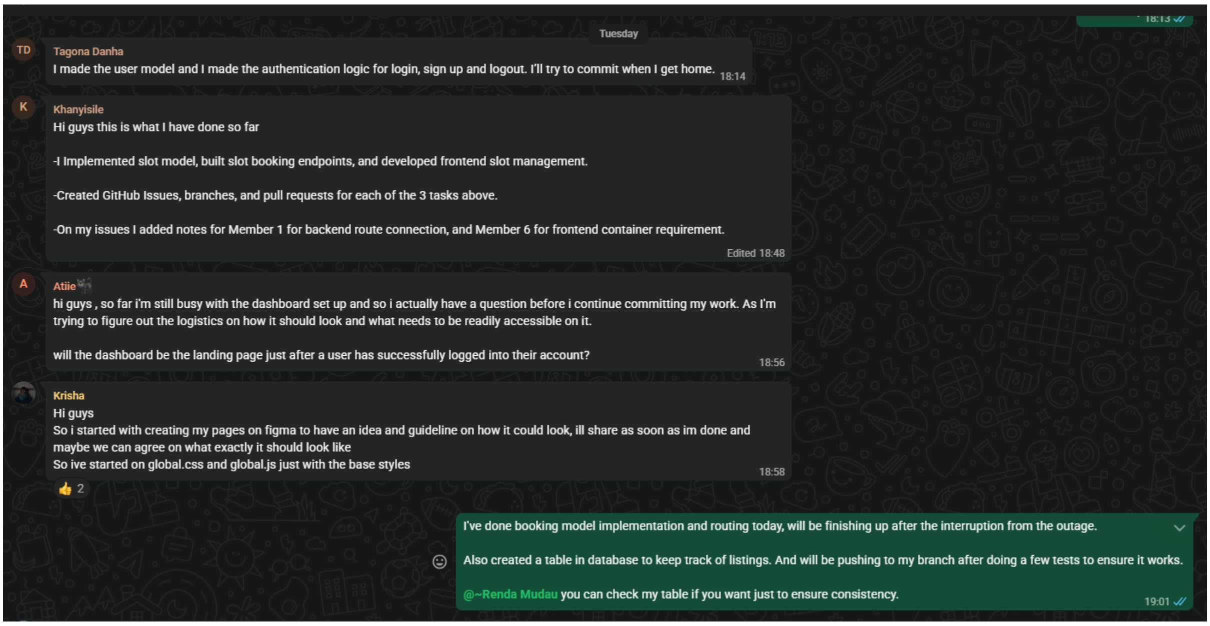
Documentation for Software Design Project

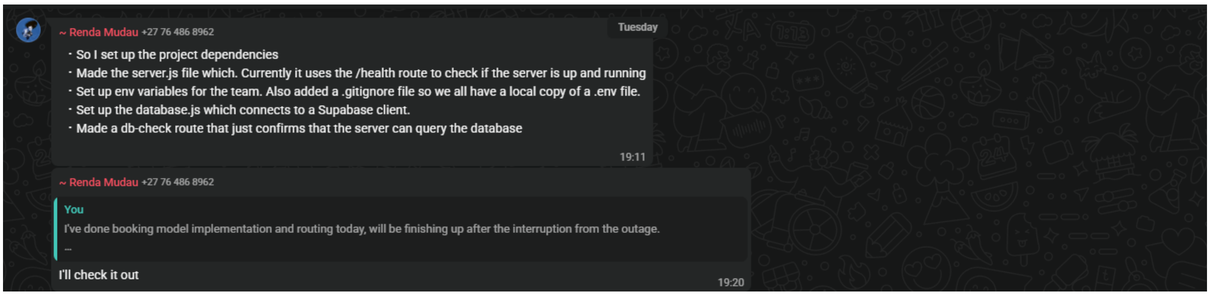
Compiled by: Zayden Suresh (2610978)

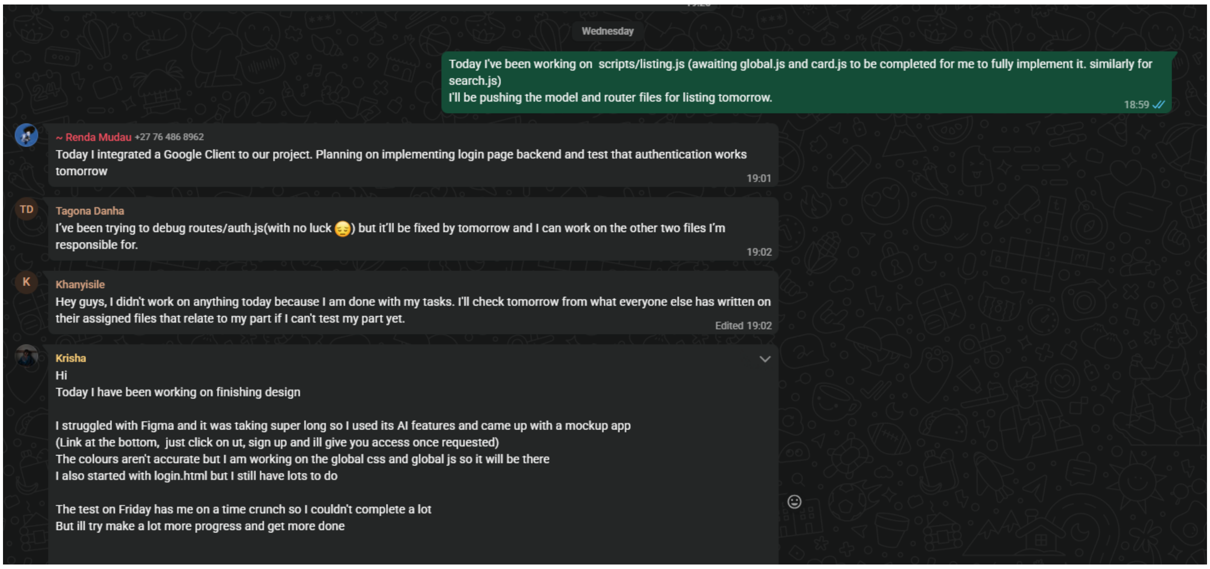
Contents

Sprint 1

Short WhatsApp meeting sessions:







Wednesday

Atiie

earlier today i implemented a few things on the dashboard, such as the navigation bar etc.. but it's not all set and done as i still need to redo the trade slots page.

also wanted to see how the login page will look at the end and take inspo from it, in order to complete the look.

Also, is one responsible in implementing code for the files that fall under "Dependencies" in their respective task or someone else is responsible for them and we just use them once they're implemented?

19:14

Thursday

Hey guys I added a line to server.js for routing listings, also added routing listings and model listings to the main repo
Search.js and Scripts listings will be done once some other files are complete

18:59

Tagona Danha

I added the user model, the routes/auth and I've written the scripts/auth

19:00

~ Renda Mudau +27 76 486 8962

I just made the initial login routing for our app. This code was all added to routes/auth.js

I made /auth/google, /auth/logout and /auth/login. I plan on testing these endpoints tomorrow. Afterwards I can begin connecting OAuth to our project to track user sessions and keep cookies of their api and stuff.

19:05

Krisha

Hi guys
I made progress with global files, I should be finishing up by tomorrow or so
Still working on some parts of login.css and html
But I should make a lot more progress tomorrow after the test 🙏

19:07

Minutes for Meeting 2026/04/04

Discussions

~11:00-11:20

First 20 minutes:

Basic greetings complete, everyone has attended

Main objectives are making a product backlog, selecting user stories for sprint 1 and assigning tasks

~11:20-11:50

Confirmed most requirements, long discussion on the meaning of "facilities". Question to tutor for clarification

~11:50-12:20

Identifying the major features, additionally adding them to the product backlog.

Must user stories be added to the backlog or must it be features? For now, features, will ask tutor later for clarification

~12:20-13:00

Product backlog selection was decided upon: Student's ability to login, student creating item listing to sell, student searching for specific items, student wanting to book a drop-off time slot to complete trades.

These are now added to Trello

Each member broke down the user stories into tasks, we have selected appropriate tasks for user stories

BREAK (13:00-13:30)

~13:30-14:15

Been working on determining dependencies of each task for optimal task allocation. Split tasks into 6 categories (for each member to select which one they want to work on)

I had split the tasks as the following (Used an LLM to assist in determining dependencies):

Member 1 – Core Backend (Server & Database Setup)
Tasks
package.json
Purpose: Define backend dependencies and scripts
Responsibilities:
• Add required dependencies (e.g., Express, CORS, dotenv)
• Define scripts (start, dev)
Dependencies: None
database.js
Purpose: Establish database connection
Responsibilities:
• Configure connection (MongoDB/MySQL)
• Handle connection errors
• Export database instance
Dependencies: None
server.js
Purpose: Main backend server setup
Responsibilities:
• Initialize Express server
• Configure middleware (JSON parser, CORS)
• Connect route files (auth, listings, slots)
• Start server
Dependencies: package.json, database.js
Potential Blockers & Solutions
• May be blocked if routes are incomplete
• Solution: Use placeholder routes so server runs early

Member 2 – Authentication & Session Management
Tasks
models/user.js
Purpose: Define user schema
Responsibilities:
• Store email and password
• Enforce unique email constraint
Dependencies: database.js
routes/auth.js
Purpose: Authentication endpoints
Responsibilities:
• Implement login, registration, and logout
• Validate input and handle errors
Dependencies: server.js, models/user.js
scripts/auth.js
Purpose: Frontend authentication logic
Responsibilities:
• Handle login/register form submission
• Send API requests to backend
• Process responses
Dependencies: Backend auth routes, global.js
scripts/session.js
Purpose: Manage user session
Responsibilities:
• Store session data (e.g., JWT) in localStorage
• Check login status on page load
• Handle logout functionality
Dependencies: auth.js, global.js
Potential Blockers & Solutions
• May depend on backend being ready
• Solution: Use mock API responses during development

Member 3 – Listings Management
Tasks
models/listing.js
Purpose: Define listing schema
Responsibilities:
• Store title, description, price, and userId
Dependencies: database.js, models/user.js
routes/listings.js
Purpose: Listings API endpoints
Responsibilities:
• Implement CRUD operations
• Associate listings with users
Dependencies: server.js, models/listing.js
scripts/listings.js
Purpose: Frontend listings logic
Responsibilities:
• Fetch and display listings
• Update and delete listings
• Implement filtering
Dependencies: Backend routes, global.js, card.js
components/search.js
Purpose: Search and filtering component
Responsibilities:
• Filter listings dynamically
• Update UI results
Dependencies: scripts/listings.js, global.js
Potential Blockers & Solutions

• May depend on authentication system
• Solution: Use temporary user data during development

Member 4 – Slots / Trade Scheduling
Tasks
models/slot.js
Purpose: Define slot schema
Responsibilities:
• Store time, availability, and booking status
Dependencies: database.js
routes/slots.js
Purpose: Slot booking endpoints
Responsibilities:
• Retrieve available slots
• Book slots
• Prevent double booking
Dependencies: server.js, models/slot.js
scripts/slots.js
Purpose: Frontend slot management
Responsibilities:
• Fetch and display slots
• Handle booking actions
Dependencies: Backend routes, global.js, card.js
Potential Blockers & Solutions
• May depend on database setup
• Solution: Use hardcoded slot data initially

Member 5 – Frontend Login & Listing Creation
Tasks
pages/login.html
Purpose: Login/register interface
Responsibilities:
• Create UI form
• Validate user input
• Display errors
Dependencies: auth.js, global.css
pages/create-listing.html
Purpose: Listing creation interface
Responsibilities:
• Build form for listing creation
• Validate input
• Submit data to backend
Dependencies: listings.js, auth.js, form.js, button.js
styles/login.css
Purpose: Login page styling
Dependencies: global.css
styles/listing.css
Purpose: Listing page styling
Dependencies: global.css
scripts/global.js
Purpose: Shared helper functions
Responsibilities:
• Create reusable API request functions (fetch wrapper)
• Handle global error handling
• Attach authentication tokens to requests
• Provide shared utility functions
Dependencies: Used by all frontend scripts
styles/global.css
Purpose: Base styling
Responsibilities:
• Define global layout (body, containers)
• Standardize fonts, colors, and spacing
• Create reusable classes (buttons, forms, cards)
• Ensure consistent UI across all pages
Dependencies: Used by all pages
Potential Blockers & Solutions
• May depend on backend and scripts
• Solution: Use placeholder handlers for UI testing

Member 6 – Dashboard & Trade Slots Frontend
Tasks
pages/dashboard.html
Purpose: Main dashboard
Responsibilities:
• Display user info
• Show listings overview
• Include navigation
Dependencies: listings.js, navbar.js, global.css
pages/trade-slots.html
Purpose: Slot booking interface
Responsibilities:
• Display available slots
• Enable booking interactions
Dependencies: slots.js, card.js, slots.css
styles/slots.css
Purpose: Slot page styling
Dependencies: global.css
Potential Blockers & Solutions

• May depend on listings and slots scripts
• Solution: Use mock data during development

Shared Resources (Used by All Members)
Components
• components/navbar.js – Navigation bar
• components/card.js – Reusable display card
• components/form.js – Form wrapper with validation
• components/button.js – Reusable button
Global Files
scripts/global.js – Shared helper functions
styles/global.css – Base styling

Renda - 1
Khanyisile - 4
TG - 2
Krisha - 5
Arnold - 6
Zayden - 3

Members have agreed tasks are reasonable

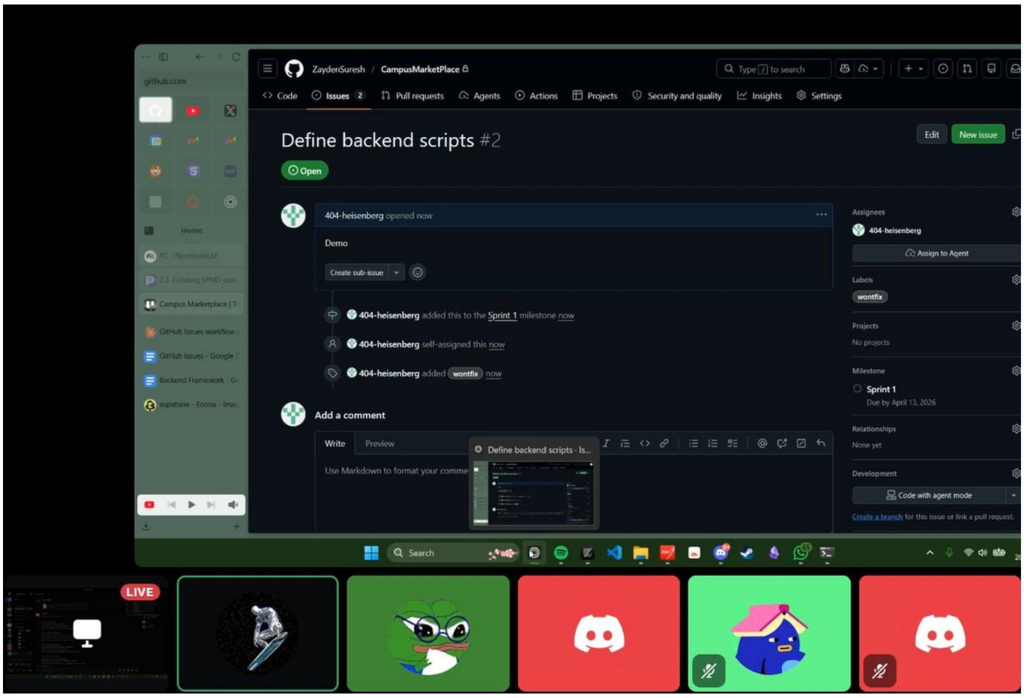
~14:15-14:30

UATs

Where will we submit UATs, when do we submit them? To be discussed with tutor.

~14:30-15:00

Will use https://github.com/ZaydenSuresh/CampusMarketPlace for project, Renda shows us how issues and branches work.



~15:00-15:10

Giving basic definition of Jest

Decided jest will be discussed more in sprint 2

CI/CD setup and Azure will also be discussed in sprint 2

~15:10-15:30

Decided we will log tasks on Trello, say what we've done on WhatsApp from Tuesday – Thursday. Have a short online meeting on Friday

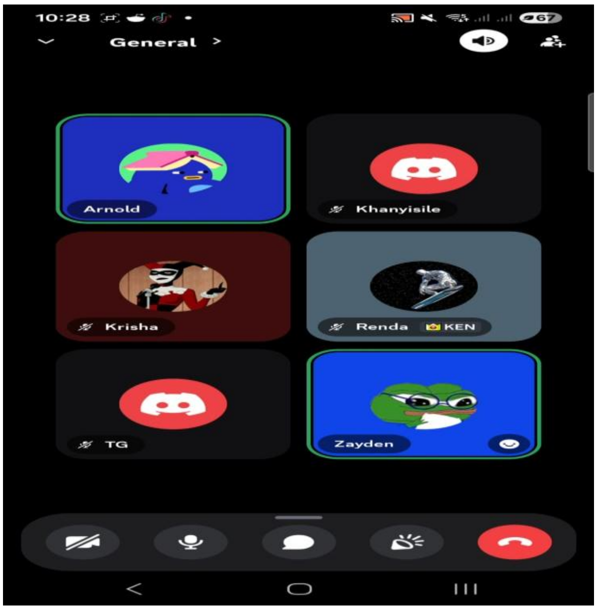
Team had questions on how tasks can be done when reliant on others (Communicate with member you are reliant on)

Decisions on task assignments reiterated, all members satisfied with their task allocation

Everyone expected to submit small bits of work throughout the week

Short closing for meeting complete

Minutes for Retrospective Meeting 2026/04/11



~10:00-10:30

Short greetings and fixing technical difficulties

Khanyisile pardoned due to medical reasons

~10:30-11:15

Each member reiterated tasks completed

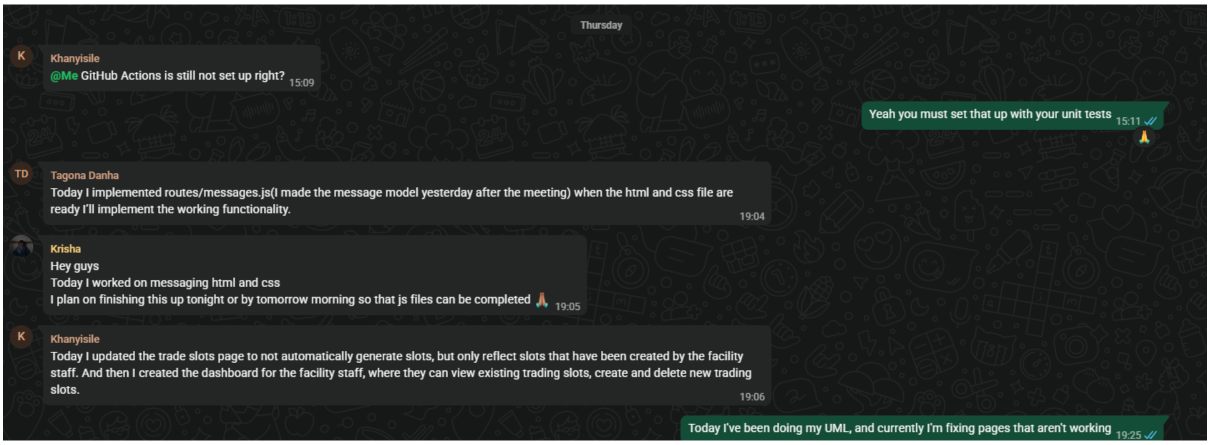
Still a few tasks needed to be completed, pushed retrospective meeting to Sunday instead

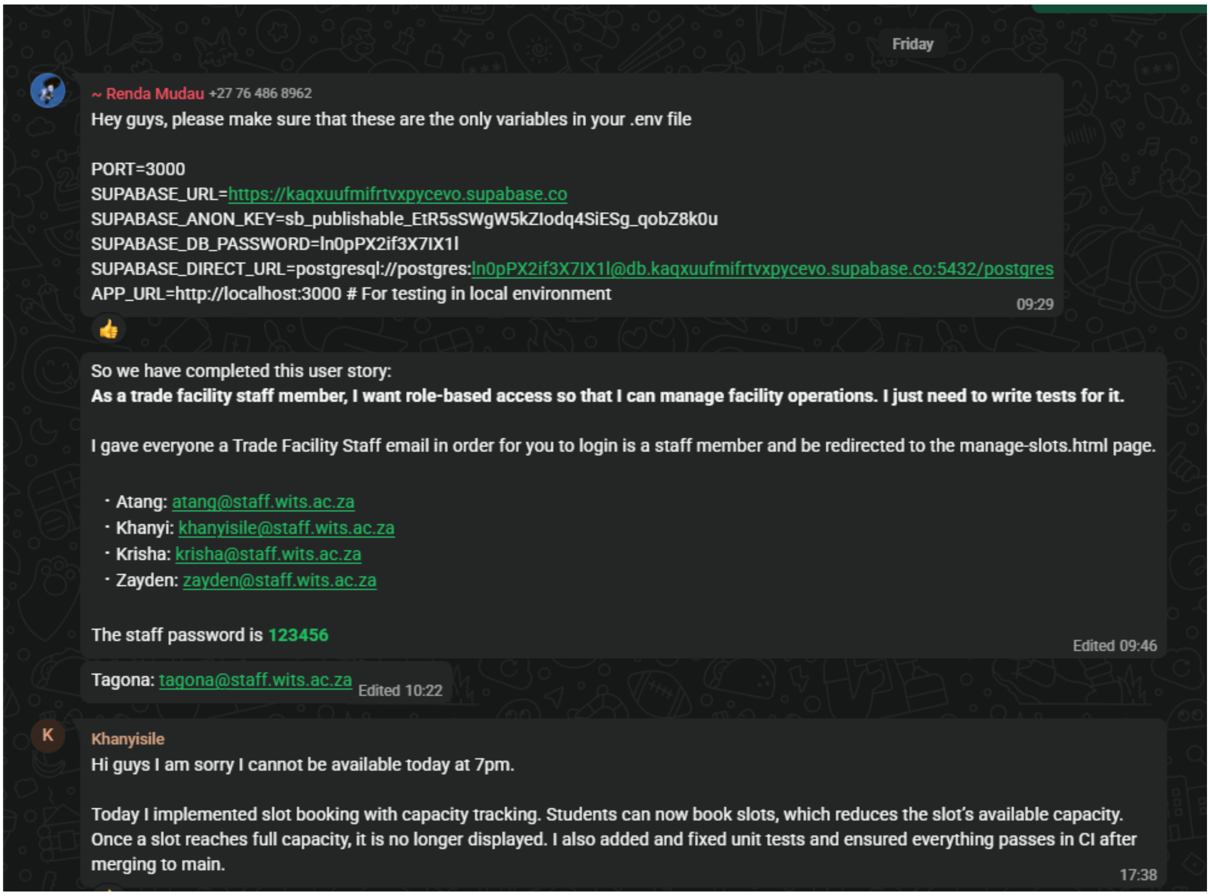
Determined inability to discuss completion of UATs and user stories due to tasks remaining

Discussed suggestions for better planning

Sprint 2:

Short WhatsApp meeting sessions





~ Renda Mudau +27 76 486 8962
Hi guys, I added my UML diagram to the repo under docs. I also finished all my tasks and just need to write the tests for them.

1. Added user role authentication. If you are a staff member you'll be taken to manage-slots.html upon sign in
2. Pages now check the role of the user when initiated. So a user who isn't a staff member can't go to manage-slots.html page and a user who isn't a student can't go to the dashboard.html page
3. I made the database filter listings rather than the app itself. Previously the database would return all listings and the app would filter them but now the listings are filtered by the database before and the app just renders them.
4. I updated scripts/auth.js to have reasuable functions like getCurrentUser, logoutUser, registerUser etc.

Edited 19:05

I'm happy to join another team if y'all need the help. I just need to write tests for what I did 19:06

Hey guys today I just did documenting of our meeting on Wednesday, I made it so that you can upload images instead of adding links to stuff on the Internet. Also added an edit button to dashboard 19:09

Krisha
Helloz
I finished the UI for messaging, all that needs to be done is the js which is currently in the works 🙏 Edited 19:10

I'll be working on my test functions mainly tomorrow, then I'll try and assist in the messaging aspect 19:11

TD Tagona Danha
> Krisha
> Helloz
> I finished the UI for messaging, all that needs to be done is the js which is currently in the works 🙏

I'm working on this. I think I'll be able to finish it by tomorrow afternoon. 19:11
🙏

A Atiie
hey guys, i'm currently still working on my UML diagram and thought the night i will be working on implementing the filter controls on dashboard.html and listings.css 19:19

Yesterday

TD Tagona Danha
> ~ Renda Mudau +27 76 486 8962
> Guys are we not meeting today? We said we'd try to get all our code done by tonight

Hey, I've made progress on the messages. 19:22

K Khanyisile
> ~ Renda Mudau +27 76 486 8962
> Guys are we not meeting today? We said we'd try to get all our code done by tonight

I think we should. 19:22

I have completed the 4th user story:

As a staff member, I want to manage booking slots so that facility capacity is controlled.

These are the headings of the user stories that I was following to implement :
Create a slot
Validate slot creation
View all slots
Edit a slot
Prevent invalid capacity update
Delete a slot
Capacity decreases when booked (control capacity)
Slot becomes unavailable when full
Staff can monitor capacity in real time
Prevent overbooking
20:34
👍

To test this, you can log in using the email address that Renda sent. 20:35

Minutes for Meeting 15/04/2026

~18:10-18:20

Reviewed sprint 2 stories and updated backlog

~18:20-18:30

I briefly discussed what UML diagrams are and the 4+1 model

I asked a question to Renda on jest

~18:30-18:40

Decided to prepare better for viva and where to put UATs

~18:40-18:50

Better team coordination has been discussed

Renda has compiled the following:

Team 1: Auth & Listings Foundation

Main Features: User authentication, role-based access control, and core listing stability

Dependencies: None (can start immediately)

Tasks:

Stabilize email login

- Wire up email/password signup and login as proper API routes so students can log in without Google.
- Files: routes/auth.js

Add role field to user model

- Extend the user model to store role (Student, Admin, Facility Staff) and return it in auth responses.
- Files: models/user.js

Implement role-based middleware

- Create middleware to protect routes so only the right roles can access certain features.
- Files: routes/*.js

Store role in session

- Save the user's role in the client-side session so the frontend knows who is logged in.
- Files: scripts/session.js

Role-based UI elements

- Show/hide navigation items and buttons based on the user's role. - Files: components/navbar.js, pages/*.html

Backend search for listings

- Move search from client-side to a proper API endpoint so it works for all users.
- Files: routes/listings.js, components/search.js

Link slot bookings to users

- Associate each booking with the user who made it so students can view their appointments.
- Files: routes/slots.js, models/slot.js

Team 2: Discovery & Listing Management

Main Features: Enhanced listing experience with filters, edit functionality, and seller information

Dependencies: Team 1 (for session/role checks once edit permissions are added)

Tasks:

Listing edit/delete

- Allow listing owners to edit or remove their own listings. - Files: routes/listings.js, scripts/listings.js

Add filter UI

- Build filter controls for category, condition, and price range.
- Files: components/search.js, pages/dashboard.html, styles/listing.css

Send filter parameters to backend

- Connect frontend filters to the search API endpoint.
- Files: scripts/listings.js, routes/listings.js

Message Seller button

- Add a button on listings that opens the messaging system.
- Files: components/card.js

Display seller rating

- Show each seller's trust rating on their listings.
- Files: components/card.js

Team 3: Transactions & Communication

Main Features: In-app messaging and trust/rating system

Dependencies: Team 1 (for user identity)

Tasks:

Create messaging model

- Set up a database table to store message history between users.
- Files: models/message.js

Messaging API routes

- Create endpoints to send messages and retrieve conversation threads. - Files: routes/messages.js

Chat UI page

- Build a page where users can view and send messages.
- Files: pages/messages.html, scripts/messages.js, styles/messages.css

Create ratings model

- Set up database table to store user ratings after transactions.
- Files: models/rating.js

Ratings API routes

- Create endpoints to submit ratings and fetch a user's average rating.
- Files: routes/ratings.js

Display ratings on dashboard

- Show seller trust scores on the main listings page.
- Files: scripts/ratings.js, pages/dashboard.html

~18:50-19:10

Brief discussion on fixing login page

Decided deadline for UML will be for Friday

Decided Renda will fix the login page

Khanyisile and Renda had a few questions

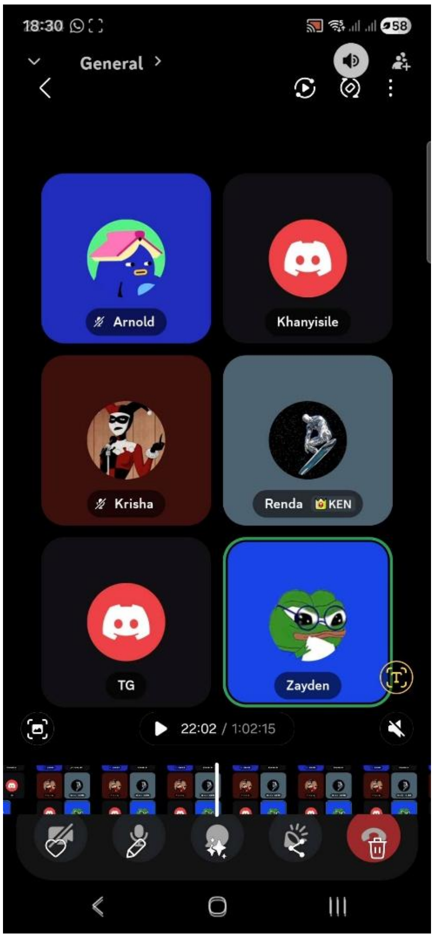
Decided on teams:

Team 1: Khanyisile and Renda

Team 2: Arnold and Zayden

Team 3: Tagona and Krisha

Meeting closed



Retrospective meeting (19/04/2026)

Written by Renda:

Sprint 2 Retrospective Meeting Minutes

1. Start and Greeting

The meeting was coordinated by Renda.

All group members were present, except for Zayden and Atang, who were absent. We greeted each other before proceeding with the session.

2. Show Work Completed

Each team member presented the work they completed during the sprint, demonstrating how their implemented user stories function within the application.

Members confirmed that:

- All assigned user stories had been completed
- Their work had been successfully pushed to GitHub
- Their features were functioning as expected when demonstrated

The group collectively reviewed the application and confirmed that the user stories for this sprint had been implemented successfully.

3. Reflect on the Sprint

3.1 Review of Previous Retrospective

The group revisited the Sprint 1 retrospective and identified that the main challenge previously was:

- Poor coordination due to members working independently on interdependent files

3.2 Improvements Implemented in This Sprint

To address this issue, the team adopted a new approach during Sprint 2 planning:

- Members were grouped into teams of two
- Each pair worked on related user stories with dependencies

3.3 Feedback on Improvements

Group members shared their experiences with this new approach:

- Collaboration within smaller teams improved coordination
- Communication between members became more effective
- Integration of dependent features was smoother

However, the group agreed that:

- Coordination can still be improved further
- Additional strategies may be needed to ensure better synchronization across all team members

4. Conclusion

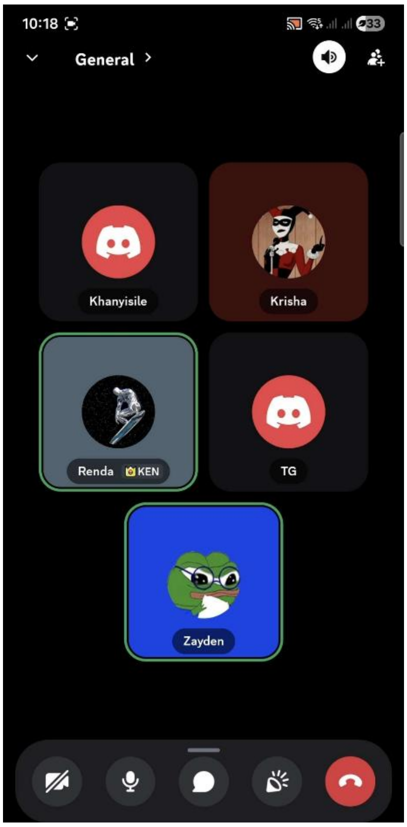
The team concluded that:

- Significant progress was made in improving coordination compared to Sprint 1
- The new team-based approach was beneficial
- Further improvements in communication and coordination will be explored in the next sprint

The meeting was then closed.

Sprint 3:

Sprint Planning Meeting Minutes – Sprint 3



Project: Campus Marketplace

Meeting Type: Sprint Planning

Date: 27 April 2026

Attendees: All team members attended

Next Meeting: 4 May 2026 (Sprint Refinement)

User Stories Selected

As a user (buyer or seller), I want to communicate through messaging before booking or completing a trade so that we can clarify and finalize transaction details.

As an admin, I want to see popular item categories so that I understand trends.

As a user, I want to rate and leave a review for another user after a transaction so that I can share my experience and help build trust within the platform.

As a student buyer, I want to purchase or reserve an item so that I can initiate a transaction with the seller.

As a student, I want to delete my listings so that I can remove unavailable items.

As a staff member, I want to confirm both item drop-off and collection so that transactions are properly tracked and completed.

Additional Tasks Discussed

After selecting the user stories, the team broke them down into smaller development tasks and discussed fixes required in the system. The following issues were identified from the sprint 2 retrospective and review with the tutor:

Fix messages not properly displaying on the application.

Improve security for the messaging feature.

Replace unnecessary <div> usage with semantic HTML.

Fix the “My Listings” button functionality.

Remove the student view button from the staff dashboard because it incorrectly redirected staff users to the student dashboard.

Remove transaction analysis from the dashboard.

Remove reserved or purchased items from the dashboard listings.

Decisions Made

User stories were finalized for Sprint 3.

Tasks were allocated among the team members.

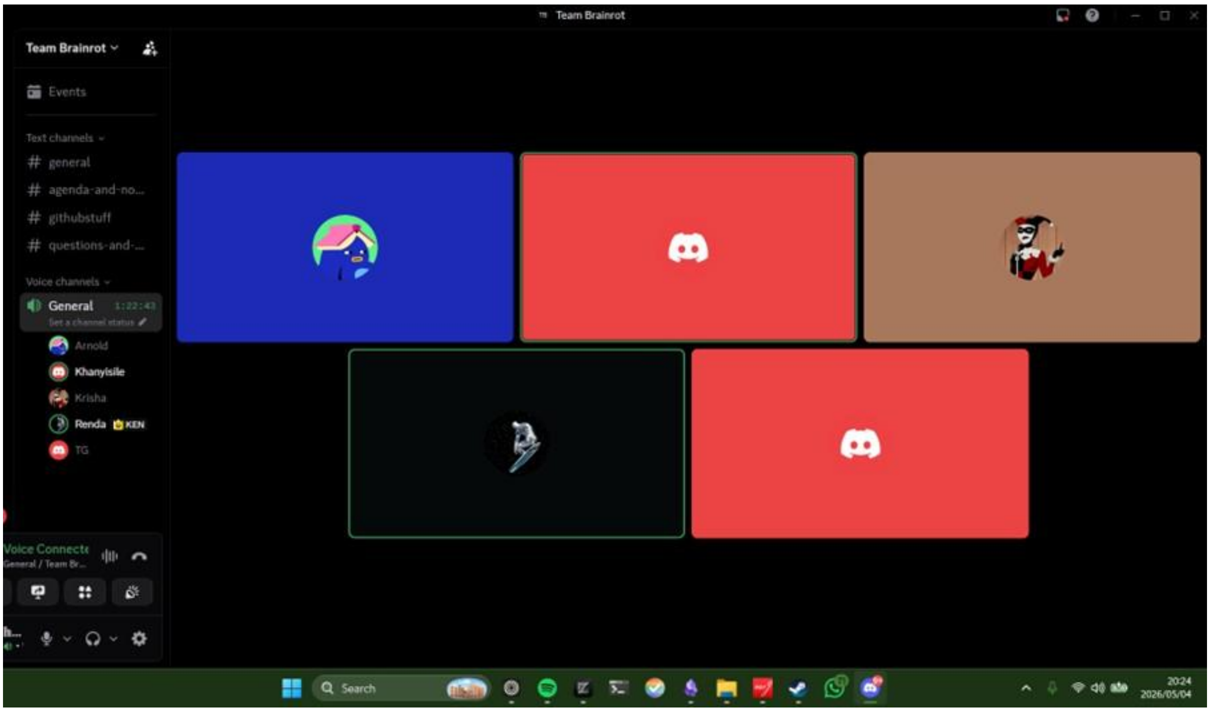
The next meeting on 4 May 2026 will focus on Sprint Refinement.

Sprint 3 Retrospective

Meeting Minutes

Monday, 04 May 2025 · 19:00

Attendance



Atang	Krisha	Tagona	Renda	Khanyisile

Sprint Updates

Summary of each team member's progress over the past two weeks.

Member	Status	Update / Notes
Renda	Completed	Delivered analytics dashboard (frontend + backend) covering all 16 transaction categories, with CSV export functionality.
Khanyisile	Completed*	Implemented Manage Trade Slots UI and full transaction workflow (Pending → Waiting Drop-Off/Pick-Up →
		Completed/Cancelled). Outstanding: separate transaction flow needed for Trades vs. Sales.
Atang	Blocked	Blocked pending Zayden's availability. Outstanding items: purchasing a listed item is non-functional; item deletion not yet started.
Tagona & Krisha	In Progress	Frontend for messaging and ratings is complete. Integration is blocked until Transactions are fully operational — ratings and meetings are inaccessible until a transaction exists in the system.

Application Workflow Discussion

The team aligned on the end-to-end user journey from item reservation through to transaction completion.

Agreed Transaction Flow

User reserves an item on the dashboard.

User receives a notification to open a conversation with the seller.

Buyer and seller negotiate and agree on a drop-off/collection time slot.

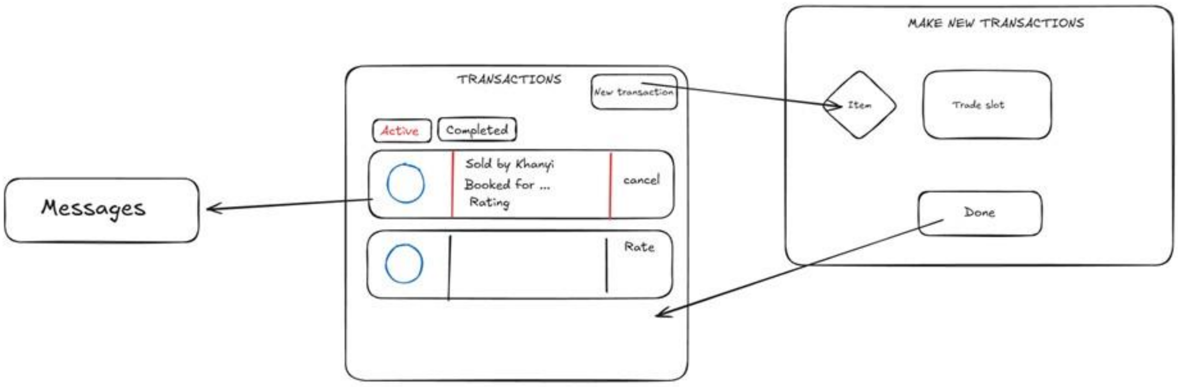
Buyer creates a new transaction, selecting a reserved item and a trade slot.

Trade facility staff tracks the transaction: Pending → In Progress → Completed.

Once completed, the buyer can rate the seller.

Key Technical Decisions

New Transactions Page required — displays all transactions for a user (identical view for buyer and seller), showing current state. Allows cancellation of active transactions and rating of completed ones.



New reservedBy column to be added to the items database table — stores the buyer's user ID upon reservation. Only one user may reserve an item at a time.

Closing

The team reflected on the value of high-level application discussions for maintaining shared understanding. All members are aligned on the target state for the sprint and the steps needed to get there.

Sprint 3 Retrospective

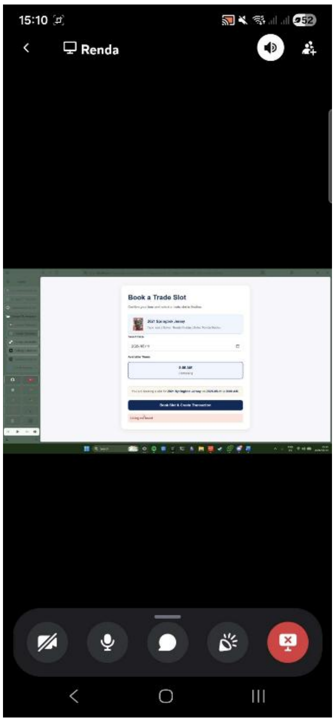
10 May 2026, 15:00

15:00-15:05:

Waiting for attendance, everyone is present

15:05-15:25

Each member discussed and presented what they did and what they have left to do



15:25:15:45

Discussed what needs to be fixed